Strong convergence of the fractional WIS Euler integrator for Lipschitz drifts

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Abstract

We prove a terminal strong convergence bound for the GfBm_EM Euler integrator of a scalar stochastic differential equation driven by multiplicative fractional Brownian noise in the Wick–Itô–Skorohod sense. For every Hurst parameter $H \in (0,1)$ and every autonomous globally Lipschitz drift, the root mean-square error is bounded by $Ch^{\min\{H+1/2,1\}}$, using exact samples of the same noise path. The drift need not have higher derivatives. Initial values may be zero, and solutions may cross zero. Conditional Gaussian averaging controls the signed quadrature error at each time scale. Summing those scales gives the fractional rate without an exponent loss. At the Brownian endpoint, time reversal and short intervals chosen by the distance from zero yield order one with polynomial localization constants.

1 Introduction

Fractional Brownian motion provides a Gaussian driving noise with correlated increments, but multiplicative equations require an interpretation of the stochastic integral. The Wick–Itô–Skorohod (WIS) interpretation permits all Hurst parameters $H \in (0,1)$ and leads to a useful exact integrating-factor representation. Erdoğan, Lord and Schieven [1] used this representation to define and implement the GfBm_EM method for scalar equations with linear multiplicative noise. Their method applies ordinary Euler time stepping to a transformed random ordinary differential equation. It is exact for the homogeneous linear equation.

For autonomous globally Lipschitz drifts, Theorem 3.6 of [1] proves a root mean-square terminal error of order h^H. The same paper conjectures the sharper rate $h^{\min\{H+1/2,1\}}$ in equation (1.3). Its Section 6 explains the difficulty: local noise defects are small, but their random coefficients prevent the cross terms from being discarded as martingale orthogonality. Summing absolute cell errors therefore loses the improvement seen in the signed time integral.

The present paper proves the conjectured autonomous estimate for the stated scheme and data class. More precisely, for every fixed $T > 0$, $H \in (0,1)$, real coefficients and initial state, and autonomous globally Lipschitz drift of at most linear growth, the terminal approximation using the exact noise samples has error

$$(\mathbb{E}|X_T - X_n|^2)^{1/2} \le C(T/n)^{\min\{H+1/2,1\}}.$$

The constant is independent of n. We require no differentiability of the drift and include zero initial states and crossings of zero. The exact recurrence and full statement are given in Section 2.

Related quadrature gains occur in several neighboring settings. Butkovsky, Dareiotis and Gerencsér [2] obtain nearly $\min\{1/2 + \vartheta H, 1\}$ strong rates for additive fractional equations under

suitable ϑ-Hölder drift conditions, using stochastic sewing. For additive equations with unbounded Lipschitz drift, Moritoki [3, Theorem 1.1] proves order one when $H > 1/2$ and order $H + 1/2 - \varepsilon$, for every $\varepsilon > 0$, when $H \le 1/2$. Altmeyer [4, Theorems 4–5] studies quadrature of occupation functionals of unperturbed fractional Brownian motion under Sobolev regularity. Kloeden and Rosa [5] establish order-one Euler convergence for semimartingale-driven random ordinary differential equations and treat a linear fractional-noise example in Appendix A. These results identify useful averaging mechanisms, but do not directly estimate the present quadrature with its random Euler state and exponential integrating factor.

Two features of the transformed equation are decisive here. Its Lipschitz constant in the state is deterministic, while the unpredictable part of that state over an interval of length ℓ is only of order ℓ^{1+H}. Conditioning in the past smooths the Gaussian noise variable. The resulting local bound retains both ℓh^H and $h\ell^H$; a dyadic martingale sum then gives $h^{H+1/2}$ for $H < 1/2$ and h for $H > 1/2$. Keeping the dependence on the interval length avoids replacing the target exponent by a nearby exponent.

The Brownian endpoint is critical for that sum. We treat it separately. Forward and backward Itô formulas give a maximal order-one quadrature bound for a Lipschitz function of Brownian motion. After separating the value of the drift at zero, the remaining noise coefficient vanishes with the transformed state. Short grid intervals of length proportional to the square of its distance from zero keep a local Girsanov density uniformly controlled. Their contributions and number have summable powers of that distance; the integrating-factor localization grows only polynomially.

Section 2 gives the exact terminal representation and common estimates. Section 3 proves the conditional fractional quadrature bound. Section 4 treats the Brownian endpoint and completes the terminal error estimate.

2 Terminal representation and common estimates

Fix $T > 0$, $H \in (0,1)$, $\alpha, \beta, x_0 \in \mathbb{R}$, and an autonomous globally Lipschitz function $a : \mathbb{R} \to \mathbb{R}$ of at most linear growth. Let B^H be centered fractional Brownian motion, normalized by

$$\mathbb{E}[B_t^H B_s^H] = \frac{1}{2}(t^{2H} + s^{2H} - |t-s|^{2H}).$$

The solution X of

$$dX_t = (\alpha X_t + a(X_t))\,dt + \beta X_t\,dB_t^H, \qquad X_0 = x_0, \tag{1}$$

uses the Wick–Itô–Skorohod (WIS) interpretation of [1]. Define, for this fixed terminal time,

$$c(t) = -\alpha t + \frac{\beta^2}{2}(T^{2H} - (T-t)^{2H}), \qquad J_t = \exp(c(t) - \beta B_t^H). \tag{2}$$

Theorem 3.2 and equations (3.6)–(3.9) of [1] give the exact representation

$$X_T = J_T^{-1} Z_T, \qquad Z_t = x_0 + \int_0^t F(c(s), B_s^H, Z_s)\,ds, \qquad F(c,b,z) = e^{c-\beta b} a(e^{-c+\beta b} z). \tag{3}$$

Thus the WIS integral is accounted for by the terminal factor and the deterministic correction in c. All subsequent time integrals involving Z are ordinary Lebesgue integrals on the same noise path.

For $n \ge 1$, put $h = T/n$, $t_i = ih$, and $\eta(t) = h\lfloor t/h \rfloor$, with endpoint values immaterial in integrals. The prescribed GfBm_EM method is

$$Z_0^n = x_0, \qquad Z_{t_{i+1}}^n = Z_{t_i}^n + hF(c(t_i), B_{t_i}^H, Z_{t_i}^n), \qquad X_n = J_T^{-1} Z_T^n. \tag{4}$$

It uses exact samples from the path appearing in (3). We use the continuous interpolation

$$Z_t^n = x_0 + \int_0^t F(c(\eta(s)), B^H_{\eta(s)}, Z^n_{\eta(s)})\, ds. \tag{5}$$

Theorem 2.1. *For every fixed data set just specified, there is a finite constant C, independent of the integer $n \geq 1$, such that the scheme (4) satisfies*

$$(\mathbb{E}|X_T - X_n|^2)^{1/2} \leq C h^{\min\{H+1/2,1\}}. \tag{6}$$

The constant may depend on T, H, α, β, x_0 and a. No derivative assumption on a, sign restriction on x_0, or restriction on $a(0)$ is imposed. The estimate concerns the terminal value and exact noise samples. Its proof appears at the end of Section 4, after the two noise regimes have been treated.

Write $\|Y\|_p = (\mathbb{E}|Y|^p)^{1/p}$. Constants C_p below may depend on the fixed problem data and on p; they are independent of n and of time variables in $[0, T]$. Let L be a Lipschitz constant for a, $a_0 = a(0)$, and $a_1 = a - a_0$. Then

$$|F(c, b, z)| \leq L|z| + |a_0|e^{c-\beta b}, \qquad |F(c, b, z) - F(c, b, z')| \leq L|z - z'|. \tag{7}$$

The second bound is deterministic despite the random integrating factor.

The dependence on the noise can also be estimated using only L. For $\sigma \in \{-1, 1\}$, define

$$b_\sigma(q) = \frac{a_1(\sigma e^q)}{\sigma e^q}. \tag{8}$$

We have $|b_\sigma| \leq L$ and $[b_\sigma]_{\mathrm{Lip}} \leq 2L$. Indeed, for $q > r$, with $x = \sigma e^q$, $y = \sigma e^r$,

$$\left|\frac{a_1(x)}{x} - \frac{a_1(y)}{y}\right| \leq 2L\frac{|x - y|}{|x|} = 2L(1 - e^{r-q}) \leq 2L(q - r).$$

For $z \neq 0$, the part of F corresponding to a_1 is

$$F_1(c, b, z) = z\, b_{\operatorname{sign} z}(\log|z| - c + \beta b), \tag{9}$$

and $F_1(c, b, 0) = 0$. Consequently,

$$\begin{aligned} |F_1(c, b, z)| &\leq L|z|, \\ |F_1(c, b, z) - F_1(c, b', z)| &\leq 2L|\beta|\,|z|\,|b - b'|, \\ |F_1(c, b, z) - F_1(c', b, z)| &\leq 2L|z|\,|c - c'|. \end{aligned} \tag{10}$$

The a_0 part is $a_0 e^{c-\beta b}$. The inequality $|e^u - e^v| \leq (e^u + e^v)|u - v|$, together with (10), gives all noise and c increments used below. For example, if z has bounded moments of the required orders, then Hölder's inequality and Gaussian exponential moments give

$$\|F(c, B^H_r, z) - F(c, B^H_u, z)\|_p \leq C_p|r - u|^H \tag{11}$$

for c in a fixed bounded interval. Independence of z and the noise is not required.

We will smooth the noise variable, rather than the original drift. Let $P_v f$ be convolution with the centered Gaussian density of variance $v > 0$, and put $f_{c,z}(b) = F(c, b, z)$.

Lemma 2.2. *For bounded c and $0 < v \le V < \infty$,*

$$|\partial_b P_v f_{c,z}(b)| \le C(|z| + e^{-\beta b}), \qquad (12)$$
$$|\partial_v P_v f_{c,z}(b)| \le C v^{-1/2}(|z| + e^{-\beta b}).$$

For $g = f_{c,z} - f_{c,z'}$,

$$\|\partial_b P_v g\|_\infty \le CL|z - z'|v^{-1/2}, \qquad \|\partial_v P_v g\|_\infty \le CL|z - z'|v^{-1}. \qquad (13)$$

Here the constants are independent of z, z', b, v in the stated ranges.

Proof. The increment estimates above bound $|f_{c,z}(b + y) - f_{c,z}(b)|$ by $C|y|(|z| + e^{-\beta b}(1 + e^{|\beta||y|}))$. Both derivatives of the Gaussian kernel have integral zero. Subtract $f_{c,z}(b)$ inside their integrals. For the first derivative, the integral of $|y|$ times its absolute value is bounded by a constant; for the second it is bounded by $Cv^{-1/2}$. The exponential weight obeys the same bounds for $v \le V$, by Gaussian scaling and exponential moments. Since $\partial_v P_v = \frac{1}{2}\partial_{bb} P_v$, this proves (12). Finally, $\|g\|_\infty \le L|z - z'|$ by (7). The L^1 norms of the first spatial and variance derivatives of the Gaussian kernel are $Cv^{-1/2}$ and Cv^{-1}, respectively, which proves (13). All differentiated integrals converge absolutely, including the exponential weights. □

Lemma 2.3. *For every finite $p \ge 1$,*

$$\sup_{n,t\le T} \|Z_t^n\|_p + \sup_{t\le T} \|Z_t\|_p \le C_p, \qquad \|Z_t^n - Z_s^n\|_p + \|Z_t - Z_s\|_p \le C_p|t - s|. \qquad (14)$$

Also J_t and J_t^{-1} have uniformly bounded moments of every finite order, and

$$\mathrm{TV}(c; [0, T]) \le |\alpha|T + \frac{\beta^2}{2} T^{2H}. \qquad (15)$$

Proof. The assertions for J follow from its Gaussian exponential formula. Equation (7) and Minkowski's inequality give the scalar Gronwall bounds for Z and the discrete Gronwall bounds for the grid values of Z^n. Equation (5) gives the same bound between grid points. Integrating the bounded L^p drift norms then gives the increment estimate. The two summands of c in (2) have total variations $|\alpha|T$ and $\beta^2 T^{2H}/2$. In particular the terminal singularity of c' when $H < 1/2$ causes no loss after integration. □

3 Conditional quadrature for fractional noise

Throughout this section $H \ne 1/2$. We realize the normalized noise using a two-sided Brownian motion W:

$$B_t^H = \gamma_H \int_{\mathbb{R}} \left((t - u)_+^{H-1/2} - (-u)_+^{H-1/2}\right) dW_u, \qquad (16)$$

where $\gamma_H > 0$ is chosen so that $\mathbb{E}(B_1^H)^2 = 1$. The integrand is square integrable: the finite singularities have squared exponent $2H - 1 > -1$, and its difference at negative infinity is $O(|u|^{H-3/2})$. The Wiener integral and its isometry are as in [6, Theorem 3.7.1]; the negative and positive half-lines use independent Brownian motions. Translation gives stationary increments and dilation gives variance t^{2H}, so polarization yields exactly the covariance in Section 2. Gaussian moment bounds and the continuity criterion of [6, Theorem 3.6.3] give a continuous version. Since the exact terminal value and its approximation in (3)–(4) are functions of the same continuous Gaussian path, this realization preserves their joint law.

Let $\mathcal{F}_s = \sigma(W_u : u \le s)$, completed, and write $\mathbb{E}^s = \mathbb{E}[\cdot \mid \mathcal{F}_s]$. This is the driving Brownian filtration, including the negative-time past; we do not identify it with the natural filtration of B^H. For $r > s$, the conditional Gaussian mean and variance are

$$m_s(r) = \mathbb{E}^s B_r^H, \qquad v_s(r) = q_H(r-s)^{2H}, \qquad q_H = \frac{\gamma_H^2}{2H}.$$

The innovation is independent of $\mathcal{F}_s$. Thus, for an $\mathcal{F}_s$-measurable z,

$$\mathbb{E}^s F(c, B_r^H, z) = P_{v_s(r)} f_{c,z}(m_s(r)). \tag{17}$$

For $s \le u < r$ with $r - u \le u - s$, the Wiener isometry and the mean-value formula for the kernel give

$$\|m_s(r) - m_s(u)\|_p \le C_p (r-u)(u-s)^{H-1}. \tag{18}$$

Indeed, the squared kernel derivative integrates on $(-\infty, s]$ to a constant times $(u-s)^{2H-2}$; Gaussian scaling gives every finite p. Moreover $\mathrm{Var}(m_s(r)) \le T^{2H}$. Its exponential moments are therefore uniformly bounded.

The slow Euler state can be predicted more accurately than the noise.

Lemma 3.1. *For $p \ge 1$ and $0 \le s < t \le T$,*

$$\|Z^n_{\eta(t)} - \mathbb{E}^s Z^n_{\eta(t)}\|_p \le C_p (t-s)^{1+H}. \tag{19}$$

Proof. The left side is zero if $\eta(t) \le s$. Otherwise put $s^+ = h\lceil s/h \rceil \le \eta(t)$. The state $Z^n_{s^+}$ is $\mathcal{F}_s$-measurable: the full Euler step ending at s^+ uses the preceding grid noise and state. The random variable

$$V = Z^n_{s^+} + \int_{s^+}^{\eta(t)} F(c(\eta(r)), B_s^H, Z^n_{s^+})\, dr$$

is $\mathcal{F}_s$-measurable. Keeping the deterministic $c(\eta(r))$ unchanged, (7), (11) and Lemma 2.3 yield

$$\|Z^n_{\eta(t)} - V\|_p \le C_p \int_{s^+}^{\eta(t)} (\eta(r) - s^+ + (\eta(r) - s)^H)\, dr \le C_p (t-s)^{1+H}.$$

Conditional contraction proves (19), with a factor two. In particular, if $s_1 \le s_2 \le r$ and $r - s_1 \le C_0 \ell$, then

$$\|\mathbb{E}^{s_1} Z^n_{\eta(r)} - \mathbb{E}^{s_2} Z^n_{\eta(r)}\|_p \le C_p \ell^{1+H}. \tag{20}$$

This follows by conditioning $Z^n_{\eta(r)} - \mathbb{E}^{s_1} Z^n_{\eta(r)}$ at s_2. □

We estimate a signed quadrature, rather than the sum of absolute cell errors. Put $c_r = c(\eta(r))$, $U_r = Z^n_{\eta(r)}$, and

$$G_r(z) = F(c_r, B_r^H, z) - F(c_r, B^H_{\eta(r)}, z), \qquad Q_h(t) = \int_0^t G_r(U_r)\, dr. \tag{21}$$

Proposition 3.2. *Uniformly for $0 \le t \le T$,*

$$\|Q_h(t)\|_4 \le \begin{cases} Ch^{H+1/2}, & 0 < H < 1/2, \\ Ch, & 1/2 < H < 1. \end{cases} \tag{22}$$

Proof. For an interval $[s,t]$ of length $\ell = t-s$, let $v = (s-\ell)\vee 0$ and define the conditional increment

$$\mathcal{A}_{s,t} = \mathbb{E}^v \int_s^t G_r(\mathbb{E}^v U_r)\, dr. \tag{23}$$

We use intervals with $s = 0$ or $s \geq \ell$, which include every dyadic interval in $[0,t]$. For every fixed $p \geq 1$,

$$\|\mathcal{A}_{s,t}\|_p \leq C_p b_h(\ell), \qquad b_h(\ell) = \min\{\ell h^H, h\ell^H\}. \tag{24}$$

The first bound is (11) integrated over the interval. For the second, suppose first that $s \geq \ell$ and $\ell \geq 2h$. Then $\eta(r) - v \geq \ell/2$. Conditional Gaussian convolution, (18), and Lemma 2.2 give, with $z_r = \mathbb{E}^v U_r$,

$$\begin{aligned}\|P_{v_v(r)} f_{c_r,z_r}(m_v(r)) - P_{v_v(\eta(r))} f_{c_r,z_r}(m_v(\eta(r)))\|_p \\ \leq C_p(h\ell^{H-1} + \ell^{-H} h\ell^{2H-1}) \leq C_p h\ell^{H-1}.\end{aligned}$$

The two terms respectively change the mean and variance. Their weighted derivative factors have bounded higher moments, by Lemma 2.3 and the Gaussian bounds on m_v. Integrating gives $C_p h\ell^H$. For $s = 0$, the part $r \leq 2h$ costs $C_p h^{1+H}$; on $r > 2h$ the same integrand estimate is $C_p h r^{H-1}$, whose integral is $C_p h\ell^H$. If $h \leq \ell \leq 2h$, the first bound implies the second up to a fixed factor; if $\ell < h$, the minimum is the first bound.

Let $u = (s+t)/2$ and $\delta\mathcal{A}_{s,u,t} = \mathcal{A}_{s,t} - \mathcal{A}_{s,u} - \mathcal{A}_{u,t}$. For $s \geq \ell$ we also have

$$\|\mathbb{E}^{s-\ell}\delta\mathcal{A}_{s,u,t}\|_p \leq C_p h\ell^{1+H}. \tag{25}$$

If $\ell \leq 4h$, the tower property expresses the conditional difference as integrals of $G_r(z_1) - G_r(z_2)$, where z_1 and z_2 are the parent and child predictions of U_r. The $2L$-Lipschitz bound on G_r and (20) give $C_p\ell^{2+H} \leq C_p h\ell^{1+H}$.

For $\ell \geq 4h$, the child's conditioning time is $w = s - \ell/2$ on the left half and $w = s$ on the right half. On its corresponding half, $\eta(r) - w \geq \ell/4$. Both $z_1 = \mathbb{E}^{s-\ell}U_r$ and $z_2 = \mathbb{E}^w U_r$ are $\mathcal{F}_w$-measurable. Apply (17) to the bounded function $g = f_{c_r,z_1} - f_{c_r,z_2}$. Equations (13) and (18) bound the conditional integrand in L^p by

$$C_p\|z_1 - z_2\|_{2p}(\ell^{-H} h\ell^{H-1} + \ell^{-2H} h\ell^{2H-1}) \leq C_p h\ell^H.$$

Integrate over the two halves and condition once more at $s-\ell$ to obtain (25). If $s = 0$, the parent and both children condition at zero, so $\delta\mathcal{A}_{0,t/2,t} = 0$ exactly.

For the dyadic subdivision of $[0,t]$ at length $\ell_j = t2^{-j}$, let S_j be the sum of its conditional increments. The difference $S_{j+1} - S_j$ is minus the sum of their $\delta\mathcal{A}$'s. At the interval $[i\ell_j, (i+1)\ell_j]$, this difference is $\mathcal{F}_{i\ell_j}$-measurable, since the latest child conditioning time is $i\ell_j$. For $i \geq 1$, subtract its conditional expectation at $\mathcal{F}_{(i-1)\ell_j}$. The resulting sequence is a martingale-difference sequence; the initial interval contributes zero. Equations (24)–(25) give

$$\|S_{j+1} - S_j\|_4 \leq C\sqrt{t/\ell_j}\, b_h(\ell_j) + Cth\ell_j^H. \tag{26}$$

For completeness, the required discrete estimate is elementary. If D_i are martingale differences, $M = \sum_i \|D_i\|_4^2$, and $S = \sum_i D_i$, then

$$S^2 = 2\sum_i S_{i-1}D_i + \sum_i D_i^2, \qquad \|S\|_4^2 \leq 2\|S\|_4\sqrt{M} + M.$$

The first sum has orthogonal L^2 increments, and conditional Jensen bounds every $\|S_{i-1}\|_4$ by $\|S\|_4$. Solving the quadratic gives $\|S\|_4 \le (1+\sqrt{2})\sqrt{M}$, which proves (26).

The sums S_j converge to $Q_h(t)$ in L^4. On an interval of length ℓ, replacing U_r by its prediction costs $C\ell^{2+H}$, by (19). Conditional projection of $F(c_r, B_r^H, z_r)$ costs $C\ell^H$ pointwise: its innovation has variance at most $C\ell^{2H}$, and the noise increment bound has integrable exponential weights. The same holds for $B^H_{\eta(r)}$, with zero cost if $\eta(r) \le v$. Hence

$$\|S_j - Q_h(t)\|_4 \le CT(\ell_j^H + \ell_j^{1+H}) \longrightarrow 0.$$

In (26), scales $\ell_j \ge h$ contribute

$$C\sqrt{t}\,h \sum_{\ell_j \ge h} \ell_j^{H-1/2},$$

whereas scales $\ell_j < h$ contribute at most $C\sqrt{t}\,h^{H+1/2}$. The conditional-mean terms sum to $Cth\sum_j \ell_j^H \le Ch$. The initial increment satisfies (24). The first geometric sum is dominated by its smallest scale if $H < 1/2$ and by its largest scale if $H > 1/2$, giving (22). For $t < h$, the direct estimate Cth^H suffices. All bounds are uniform for $t \le T$. □

At $H = 1/2$, the first geometric sum is critical and this argument would give $h(1+\log(T/h))$. We next use the Brownian structure to recover order one without differentiating the drift.

4 The Brownian endpoint

Here $H = 1/2$, $B^{1/2} = W$ is ordinary Brownian motion, and $c(t) = (-\alpha + \beta^2/2)t$. We first record the quadrature estimate that replaces the critical scale sum. The stochastic-calculus facts used in this section are ordinary Itô calculus, the Brownian martingale characterization, Girsanov's theorem for bounded progressive drifts, and the Burkholder–Davis–Gundy inequalities; see [6, Theorems 7.1.1, 7.2.1, 7.5.1 and 7.7.1].

Lemma 4.1. *Let $W_0 = 0$, $d = mh$ for an integer $m \ge 1$, and $f : \mathbb{R} \to \mathbb{R}$ be globally Lipschitz. For every $p \ge 2$,*

$$\left\| \sup_{t \le d} \left| \int_0^t (f(W_s) - f(W_{\eta(s)}))\, ds \right| \right\|_p \le C_p [f]_{\mathrm{Lip}} h\sqrt{d}. \tag{27}$$

The bound is independent of $f(0)$ and of shifts of its argument.

Proof. First suppose f is smooth with bounded first derivative. Set $g_h(s) = (\lfloor s/h \rfloor + 1)h - s$. At grid endpoints the quadrature is $\int_0^t g_h(s)\, df(W_s)$. At intermediate t, its difference from that integral is

$$-((\lfloor t/h \rfloor + 1)h - t)(f(W_t) - f(W_{\eta(t)})).$$

The maximum of this correction has L^p norm at most $C_p[f]_{\mathrm{Lip}} h\sqrt{d}$.

The change $df(W)$ is the average of its forward and right-endpoint backward stochastic differentials:

$$\int_0^t g_h\, df(W) = \frac{1}{2}\int_0^t g_h(s) f'(W_s)\, dW_s + \frac{1}{2}\int_0^t g_h(s) f'(W_s)\, d^- W_s. \tag{28}$$

The forward integral equals the change in $f(W)$ minus one-half the f'' integral, while the backward integral equals that change plus one-half the f'' integral. The deterministic bounded-variation

function g_h adds no quadratic covariation. This proves (28) first on each cell and then on their union.

To estimate the backward integral, let $Y_u = W_{d-u}$ in its natural reverse filtration. Its Gaussian bridge transition gives, for $u < v < d$,

$$\mathbb{E}[Y_v \mid Y_r,\ r \le u] = \frac{d-v}{d-u} Y_u.$$

It follows that $M_u = Y_u - Y_0 + \int_0^u Y_r/(d-r)\,dr$ is a continuous martingale. Its quadratic variation is u, because the added integral has finite variation and Brownian increments retain their quadratic variation on reversal. The Brownian martingale characterization therefore gives

$$dY_u = dM_u - \frac{Y_u}{d-u}\,du.$$

The drift extends to the endpoint in every finite L^p, since $\int_0^d \|Y_u\|_p/(d-u)\,du = 2C_p\sqrt{d}$. Writing $N_u = \int_0^u g_h(d-v)f'(Y_v)\,dM_v$, the reversal is explicitly

$$\int_0^t g_h(s)f'(W_s)\,d^-W_s = -\int_{d-t}^d g_h(d-u)f'(Y_u)\,dY_u = -(N_d - N_{d-t}) + \int_{d-t}^d g_h(d-u)f'(Y_u)\frac{Y_u}{d-u}\,du.$$

The martingale increment is controlled by $2\sup_{u\le d}|N_u|$; its integrand is bounded by $h[f]_{\mathrm{Lip}}$. The drift term is bounded by

$$h[f]_{\mathrm{Lip}} \int_0^d \frac{|W_s|}{s}\,ds,$$

whose L^p norm is at most $2C_p h[f]_{\mathrm{Lip}}\sqrt{d}$. The forward martingale maximum has the same bound by the maximal martingale inequality. One can first reverse before d and then pass to d using the displayed integrability bounds. This proves (27). Finally, mollify f by a Gaussian kernel. The Lipschitz constant does not increase and the approximations converge uniformly, so their quadratures converge uniformly. The same estimate holds for f. □

Lemma 4.2. *Suppose $\beta \neq 0$ and, on a grid block of length $d = mh$,*

$$dq_t = u(q_t)\,dt + \beta\,dW_t, \qquad |u| \le M.$$

If $M^2 d \le C_0$, then for every bounded Lipschitz b and $p \ge 2$,

$$\left\| \sup_{t\le d} \left| \int_0^t (b(q_s) - b(q_{\eta(s)}))\,ds \right| \right\|_p \le C_{p,C_0,\beta}[b]_{\mathrm{Lip}} h\sqrt{d}. \tag{29}$$

The constant is uniform in the initial value. The same estimate holds conditionally at a stopping time in a finite deterministic grid, with the initial value and block length known at that time.

Proof. Let $\mu_t = u(q_t)/\beta$ and $D = \exp(-\int_0^d \mu_t\,dW_t - \frac{1}{2}\int_0^d \mu_t^2\,dt)$. The bounded progressive coefficient makes this a probability density. Under $d\mathbb{Q} = D\,d\mathbb{P}$, the process $\widetilde{W}_t = W_t + \int_0^t \mu_s\,ds$ is Brownian and $q_t = q_0 + \beta\widetilde{W}_t$. For each fixed moment exponent, both the forward and inverse densities have moments bounded by $\exp(C_p M^2 d/\beta^2)$; this follows by factoring their powers into an exponential martingale and its bounded quadratic-variation correction. Hölder's inequality and Lemma 4.1, applied at exponent $2p$ to $f(x) = b(q_0 + \beta x)$, prove (29). At a finite-grid stopping time, condition on each of its possible values. The future Brownian increments are independent of the past, and all block parameters are already known. The same density argument gives the conditional assertion. □

We retain the original Z-equation for stability. The logarithm will only be used on intervals where Z stays away from zero. On a sign interval $\sigma = \operatorname{sign} Z$, recall

$$b_\sigma(q) = \frac{a(\sigma e^q) - a_0}{\sigma e^q}, \qquad Z_t' = J_t a(Z_t/J_t).$$

For $q_t = \log|Z_t/J_t| = \log|Z_t| - c(t) + \beta W_t$, the finite-variation term has differential $(Z_t'/Z_t)\,dt$. Since $c' = -\alpha + \beta^2/2$, this gives

$$dq_t = \left(\alpha - \frac{\beta^2}{2} + b_\sigma(q_t) + \sigma a_0 e^{-q_t}\right) dt + \beta\, dW_t. \tag{30}$$

If $a_0 = 0$ and $x_0 \neq 0$, then $|x_0|e^{-LT} \le |Z_t| \le |x_0|e^{LT}$, so its sign never changes. The drift in (30) is bounded. The noise quadrature vanishes when $\beta = 0$, so assume $\beta \neq 0$ here. Define

$$R(t) = \int_0^t [b_\sigma(q_s) - b_\sigma(q_{\eta(s)})]\, ds.$$

Lemma 4.2 over the whole interval gives $\|\sup_{t\le T}|R(t)|\|_p \le C_p h$. By (7), (10) and Lemma 2.3, the exact-state F_1 noise quadrature on a grid prefix, defined in (31) below, differs from $\int_0^{t_k} Z_{\eta(s)}\, dR(s)$ by $O(h)$ in L^p. Summation by parts bounds this weighted integral by

$$\Big(\sup_{t\le T}|Z_t| + \mathrm{TV}(Z_\eta;[0,T])\Big)\sup_{t\le T}|R(t)|.$$

The first factor is deterministically bounded, so the F_1 noise quadrature has order h. If also $x_0 = 0$, the solution and scheme vanish identically.

When $a_0 \neq 0$, the last drift term in (30) can be singular. Short blocks chosen by the distance of Z from zero keep its density cost bounded. The resulting estimates grow only polynomially in the integrating factor.

Proposition 4.3. *For $H = 1/2$, uniformly at grid times t_k,*

$$\left\| \int_0^{t_k} \big(F_1(c(\eta(s)), W_s, Z_{\eta(s)}) - F_1(c(\eta(s)), W_{\eta(s)}, Z_{\eta(s)})\big)\, ds \right\|_4 \le Ch. \tag{31}$$

Proof. The case $a_0 = 0$ was just discussed, with the $O(h)$ state and c freezing bounds from (10) and Lemma 2.3. If $\beta = 0$, the integral vanishes. Reflection $X \mapsto -X$ permits us to assume $a_0 > 0$. If $L = 0$, then $a_1 = 0$, so suppose $L > 0$. Let

$$M_J = \max\{1, \sup_{t\le T} J_t, \sup_{t\le T} J_t^{-1}\}, \qquad E_K = \{M_J \le K\}, \quad K \ge 2.$$

On E_K, deterministic Gronwall gives

$$\sup_{t\le T}|Z_t| \le C_Z K, \qquad \sup_{t\le T}|Z_t'| \le C_D K. \tag{32}$$

Choose a fixed $r_* > 0$ with $r_* \le \min\{1, a_0/(4L)\}$ and put $r_0 = r_*/K$. In the band $|Z_t| \le 2r_0$,

$$Z_t' = a_0 J_t + F_1(c(t), W_t, Z_t) \ge \frac{a_0}{2K}. \tag{33}$$

Thus Z traverses the band monotonically. A positive or negative annulus inside the band is traversed at most once; the time spent in $\epsilon/2 \le |Z| \le 3\epsilon$ is at most $CK\epsilon$ when $3\epsilon \le 2r_0$.

Choose a sufficiently large fixed $D \geq 8C_D C_Z + 1$, and put $\delta = 2\sqrt{D}\,K\sqrt{h}$. We first assume $h \leq c_* K^{-4}$, where the fixed $c_* > 0$ is small enough that $2\delta \leq r_0$ and $C_D K h \leq \delta$. Partition a prefix $[0, t_k]$ at grid times as follows. A start t_i with $|Z_{t_i}| \leq \delta$ receives one cell. Otherwise choose $\epsilon = \delta 2^j$ with $\epsilon \leq |Z_{t_i}| < 2\epsilon$, and take

$$m_i = \min\left\{k - i, \left\lfloor \frac{\epsilon^2}{DK^2 h} \right\rfloor\right\} \tag{34}$$

consecutive cells. Every start is a finite-grid stopping time. Its sign, ϵ, and length $d = m_i h$ are known at that time. The floor is at least four before the final truncation, and

$$d \leq \frac{\epsilon^2}{DK^2}, \qquad d \geq \frac{\epsilon^2}{2DK^2}$$

unless this is the last block.

On E_K, $\epsilon \leq C_Z K$, and (32) gives $|Z_t - Z_{t_i}| \leq C_D K d \leq \epsilon/4$. Throughout an outer block, Z therefore keeps its sign and $\epsilon/2 \leq |Z| \leq 3\epsilon$, while $|Z/J| \geq \epsilon/(2K)$. Clip the last term of (30) to

$$\sigma \min\{a_0 e^{-q}, 2a_0 K/\epsilon\}.$$

This produces a globally Lipschitz bounded drift agreeing with the exact q on the block on E_K. For starts with $\epsilon \leq C_Z K$, its bound is CK/ϵ, and its squared bound times d is uniformly bounded. Let q^{cl} be this clipped process with $q^{\mathrm{cl}}_{t_i} = q_{t_i}$. For this block $b = [t_i, t_i + d]$, put

$$R^{\mathrm{cl}}_b(u) = \int_{t_i}^{u} [b_\sigma(q^{\mathrm{cl}}_s) - b_\sigma(q^{\mathrm{cl}}_{\eta(s)})]\, ds, \qquad t_i \leq u \leq t_i + d.$$

On E_K, write R_b for the same primitive using the unclipped q; the supremum below is over this block. Lemma 4.2, with test function b_σ, gives conditionally at the start

$$\left(\mathbb{E}\left[\sup|R^{\mathrm{cl}}_b|^p \mid \mathcal{F}_{t_i}\right]\right)^{1/p} \leq C_p h\sqrt{d} \leq C_p h\epsilon/K. \tag{35}$$

The Lipschitz constant of b_σ is at most $2L$, independently of the clipping level.

On this block, the integrand in (31) differs from $Z_{\eta(s)}[b_\sigma(q_s) - b_\sigma(q_{\eta(s)})]$ by at most CKh on E_K. Indeed, (10) and (32) freeze Z_s and the linear $c(s)$ at cost CKh. These errors over all blocks cost CKh. Summation by parts bounds the weighted quadrature by

$$\big(\sup|Z| + \mathrm{TV}(Z_\eta; \mathrm{block})\big)\sup|R_b| \leq C\epsilon \sup|R_b|,$$

since $\mathrm{TV}(Z_\eta; \mathrm{block}) \leq C_D K d \leq \epsilon/4$. On E_K the original and clipped processes agree. Thus each block contribution, multiplied by $\mathbf{1}_{E_K}$, has L^p norm at most $C_p h\epsilon^2/K$.

For $\epsilon < r_0/4$, every such block lies in its expanded annulus. Equations (33) and (34) bound the number of these disjoint blocks by CK^3/ϵ, apart from one final truncated block. Their total L^p contribution is bounded by

$$C_p h K^2 \epsilon + C_p h\epsilon^2/K.$$

Summing the dyadic ϵ's below $r_0/4$ gives a polynomial multiple of h. For the remaining outer blocks, $\epsilon \geq r_0/4$; every nonfinal length is at least cK^{-4}. There are at most CK^4 of them, and each contribution is at most $C_p hK$ because $\epsilon \leq C_Z K$ on E_K. They therefore cost at most $C_p hK^5$.

These counts remain valid in the norm estimates even though the block starts are random. For a fixed small annulus, enumerate its successive starts as grid stopping times. On E_K, only the first deterministic upper bound on their number is present. Each contribution with its indicator is

bounded by $C\epsilon \sup|R_b^{\mathrm{cl}}|$; (35) gives its norm conditionally at that start. For large blocks enumerate only starts with $\epsilon \le C_Z K$, a condition already known at the start. On E_K this includes every large block. Apply the deterministic count bound and Minkowski's inequality. Missing starts contribute zero at t_k. No independence of E_K and the block errors is used.

It remains to bound the cells starting within $|Z_{\eta(s)}| \le \delta$. On E_K, all their times lie within $|Z| \le 2\delta$, by $C_D K h \le \delta$. Their total length is at most $CK\delta$. The noise increment estimate (10) bounds their quadrature pathwise by

$$C\delta(K\delta)D_h \le CK^3 h D_h, \qquad D_h = \max_{i<n} \sup_{s\in[t_i,t_{i+1}]} |W_s - W_{t_i}|.$$

The Brownian maximal tail bound and a union bound give $\|D_h\|_p \le C_p\sqrt{h\log(2n)}$, uniformly bounded as $h = T/n$. This inner contribution is consequently at most $C_p h K^3$ in L^p. Writing $Q_h^{(1)}(t_k)$ for the integral in (31), we have proved

$$\|Q_h^{(1)}(t_k)\mathbf{1}_{E_K}\|_p \le C_p h K^6, \qquad p \ge 2. \tag{36}$$

The harmless enlargement from K^5 to K^6 includes all preceding constants. If $h > c_* K^{-4}$, the direct increment and moment bounds give $\|Q_h^{(1)}(t_k)\|_p \le C_p\sqrt{h} \le C_p h K^2$, so (36) holds for every n.

Finally $J_t = e^{c(t)-\beta W_t}$, and the Brownian supremum tail implies

$$\mathbb{P}(M_J > K) \le C\exp\{-c_1(\log K - C_1)^2\}$$

for large K, with fixed constants. For dyadic K, Hölder's inequality and (36) at $p = 8$ give

$$\|Q_h^{(1)}(t_k)\mathbf{1}_{\{K/2<M_J\le K\}}\|_4 \le ChK^6\mathbb{P}(M_J > K/2)^{1/8}.$$

The dyadic sum is finite, independently of n, proving (31). □

The constant drift part is $a_0 J_t$. Ordinary Itô calculus gives

$$dJ_t = (-\alpha + \beta^2)J_t\,dt - \beta J_t\,dW_t. \tag{37}$$

Integrating this differential with the deterministic sawtooth weight bounded by h, and using Gaussian exponential moments and the maximal martingale inequality, shows

$$\left\| \int_0^{t_k} (J_s - J_{\eta(s)})\,ds \right\|_4 \le Ch. \tag{38}$$

Combining (31) and (38) therefore controls the entire Brownian drift quadrature.

Proof of Theorem 2.1. Let $e_k = Z_{t_k} - Z_{t_k}^n$ and $\rho = \min\{H + 1/2, 1\}$. For $H \ne 1/2$, subtract (5) from (3) and use the decomposition

$$\begin{aligned} F(c(s), B_s^H, Z_s) - F(c(\eta(s)), B_{\eta(s)}^H, Z_{\eta(s)}^n) &= F(c(s), B_s^H, Z_s) - F(c(s), B_s^H, Z_{\eta(s)}^n) \\ &\quad + F(c(s), B_s^H, Z_{\eta(s)}^n) - F(c(\eta(s)), B_s^H, Z_{\eta(s)}^n) \\ &\quad + G_s(Z_{\eta(s)}^n). \end{aligned}$$

The first line costs $Ch + Lh\sum_{i<k}\|e_i\|_4$, by (7) and Lemma 2.3. The c increments cost $Ch\,\mathrm{TV}(c)$: their L^4 weighted factors are uniformly bounded, and

$$\int_0^{t_k} |c(s) - c(\eta(s))|\,ds \le h\,\mathrm{TV}(c).$$

Proposition 3.2 bounds the last line by Ch^ρ. Hence

$$\|e_k\|_4 \le Ch^\rho + Lh\sum_{i<k}\|e_i\|_4. \tag{39}$$

For $H = 1/2$, use instead the exact-state residual

$$R_k = \int_0^{t_k} \left(F(c(s), W_s, Z_s) - F(c(\eta(s)), W_{\eta(s)}, Z_{\eta(s)})\right) ds.$$

Freezing Z_s and the linear $c(s)$ costs Ch. The remaining a_1 quadrature is Proposition 4.3; the a_0 quadrature is (38). Thus $\|R_k\|_4 \le Ch$. The deterministic z-Lipschitz estimate at each grid point gives (39) with $\rho = 1$. The case $\beta = 0$ follows directly from deterministic Euler stability and consistency.

Discrete Gronwall yields $\|e_n\|_4 \le Ch^\rho$. The prescribed quotient uses the same strictly positive J_T for the exact and approximate terminal states. Therefore

$$\|X_T - X_n\|_2 \le \|e_n\|_4 \|J_T^{-1}\|_4 \le Ch^\rho,$$

as required. These estimates may first be used for $h \le \min\{T, 1\}$. The finitely many larger steps have finite errors by the moment bounds, and enlarging the fixed-data constant includes them. If $a = 0$, both transformed states are identically x_0 and the scheme is exact. □

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